

Laboratory Work Table of Contents

CSC316: Cryptography

SN	Laboratory Title (Complete Sentence)	Date	Remarks (Teacher Signature)
1	Implement the Caesar cipher encryption and decryption program.	_____	_____
2	Implement the Hill cipher encryption and decryption program for a given key matrix.	_____	_____
3	Implement the Vigenère cipher encryption and decryption program.	_____	_____
4	Implement the Vernam cipher encryption and decryption program using XOR-based logic.	_____	_____
5	Implement the One-Time Pad (OTP) encryption and decryption program with a truly random key of equal length.	_____	_____
6	Implement the Playfair cipher encryption and decryption program using a keyword-based 5×5 matrix.	_____	_____
7	Implement the Rail Fence transposition cipher encryption and decryption program for a given key (rails).	_____	_____
8	Demonstrate the working of a DES S-Box by performing substitution on a 6-bit input and producing a 4-bit output.	_____	_____
9	Implement the DES key generation steps to generate subkeys for multiple rounds (at least for initial rounds).	_____	_____
10	Implement modular arithmetic operations and compute additive inverses in Z_n for given inputs.	_____	_____
11	Implement the Extended Euclidean Algorithm to compute gcd and multiplicative inverse in Z_n when it exists.	_____	_____
12	Implement primality testing for a number using Fermat test and/or Miller–Rabin test.	_____	_____
13	Implement Euler’s totient function computation and verify Euler’s theorem for selected values.	_____	_____
14	Implement a primitive root checking program and find primitive roots for a given prime modulus.	_____	_____

SN	Laboratory Title (Complete Sentence)	Date	Remarks (Teacher Signature)
15	Implement the Diffie–Hellman key exchange and demonstrate how both parties derive the same shared secret key.	_____	_____
16	Implement the RSA algorithm (key generation, encryption, and decryption) for small sample values and text mapping.	_____	_____
17	Implement the ElGamal cryptographic system (key generation, encryption, and decryption) for sample messages.	_____	_____
18	Demonstrate basic logic of malicious code by simulating a simple harmless scanner/replication behavior in a controlled way (no harm).	_____	_____